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LOCAL SUPPORT, TOTAL COMMITMENT

«Local support, Total commitment »

As you have noticed, our TLD logo has been slightly relooked and rejuvenated at the end of 2011

But more importantly, we have created a new tagline enhancing our logo: Local support, Total commitment.

Local support is what our customers expect from us, more and more and everywhere. The expectation and standard for customer support, which used to be very different from one place to another some years ago, are now converging around the world, as all customers request a professional, diligent and efficient support for their GSE.

Total commitment means, as our first group value says, that we stand behind our customers and products, that responsibility, support and continuous improvement are the key words which drive our work and attitude.

This tagline marks our goal and ambition.

But in order to make it 100% legitimate, there must be a reality beyond the claim, built by our spirit and daily behaviour and experienced and recognized by our customers.

At TLD, we believe more than ever that this direction is key for our success.

Thank you for contributing every day to make this tagline a reality.

Jean-Marie FULCONIS

The latest dispatch

Taxibot takes off!

Originally launched 2 and half year ago, the Taxibot tow tractor concept jointly designed by IAI and TLD is now nearing production.

The Taxibot concept allows to tow aircraft from the boarding bridge to the entry of the runway with its engines shut down. After a classic push back like with any other tractor, the pilot takes control of the Taxibot and drives the tractor from the cockpit with the tiller and the airplane brakes. The main benefits of the Taxibot are the fuel savings for the airlines as well as the lower CO2 and noise emissions, that will make the airport environment greener.

The Taxibot concept has been successfully tested by demonstrator built by IAI and a special trailer in UK. It has been validated by tests on Airbus A-340 in Toulouse (France) and also on severe trials on Boeing B747 in winter 2011 at Frankfurt airport (Germany). In parallel, TLD has designed and assembled the first Narrow Body prototype which has been awarded the price of "innovation of the year" at the Interairport Show Munich in October 2011.

After the integration of the software specific to towbarless tractors on the unit, the first autonomous displacements have been made in the Montlouis factory. But the 600 kW powered unit, with a complete electric drive line, requires a lot of space for test and dynamic validation, repeated acceleration and braking tests.

Therefore, it is in Châteauroux Airport, relatively close to Montlouis, that an Airbus A 320 specifically rented for this purpose, is now waiting for the Taxibot in order to achieve dynamic test. In April 2012 we will test on this real aircraft all the solutions and innovations developed by our partners and by TLD.

The integration of the complex electronic systems developed by our partner is now in process and include anti-collision detection system, high precision GPS, yaw sensor, and a full advanced traction control. All this systems are based on aviation technology, and it is a huge challenge to make all wirings and integration with such demanding standards.

In parallel TLD is developing the Wide Body version of the TaxiBot, with 1000 kW, 12 wheels, and a capacity for taxiing up to the A 380. Our goal is to have the prototype on its wheels for the first quarter 2013.

Because of the high level of safety required for aircraft taxiing operation, Taxibot will be partially certified with an equivalent level of requirement and through the same process as an aircraft. Besides the usual certification complexity, we will also need to be patient as the certification process should last until the end of 2013.

We are getting ready to assemble the first production units of the Narrow body TaxiBot and hope to be able to make "real life" demonstration of the machine in a major airport within one year from now.

TLD also have some involvement in marketing and service subjects, so Airbus, IAI and TLD are working closely together to be ready to offer a complete solution to the end customer.

Taxibot is heading for take off..!



Hervé CRIQUILLION

The Life digest :

3

TLD aircraft tractor for the US Navy.

After 2 full years of development testing and production preparation, TLD-Canada has entered the production phase for the TMX-150-35 for its customer the United States Navy (USN). Since January 2012, the USN has been taking delivery of 4 units per month and is scheduled to do so for a total of 174 tractors. Starting in July the production rate will increase to 6 units per month.

Also known under the USN Program designation as the LLTT, (Large Land based Tow Tractor) this production program is the culmination of a challenging qualification program where 5 prototype tractors were subjected to Military type environmental and reliability testing including being exposed to Arctic cold and Desert hot conditions, blowing sand and rain and shock and vibration associated with air transport. The requirements were so extensive to include the creation of a 800 meter oval driving track where the 5 prototypes were required to perform 1500 hours of endurance testing by towing a sophisticated dynamometer that represented the loads associated with the aircraft fleet to be towed.

This qualification testing was followed by field testing by Navy personnel to insure all aspects of the tractor were perfect before proceeding with production. This test program was certainly the most intensive of any TLD product to date and was a key learning experience necessary for TLD's success as we enter more Military programs. This testing also validated the basic TMX-250 design, which the LLTT is based on, as being an extremely reliable and rugged tractor, more than ready for the commercial market.

This challenging program and especially this production phase is bringing a new dimension to TLD's ability to market the mid size conventional tractor line in the Americas by firmly establishing TLD-Canada as a manufacturer in the region. This important customer reference will no doubt benefit the entire TLD group in growing the reference of the TMX-250/350 class of tractors around the world and providing increased volumes of key components with which to leverage cost benefits with our suppliers.

In fact, today, this new production line is providing a continuous ability for TLD-Canada to display this product class to all of its prospective customers whenever they visit the factory and provides objective evidence of the product's viability and of the factory's ability to support it in the region. Since the completion of the USN qualification program several customers have already purchased or are showing strong interest in the commercial versions which will soon be insuring that a regular flow of commercial tractors will be built alongside the USN version, establishing TLD-Canada as a major player in the mid size conventional tractor product line.



LLTT towing the USN P-3 Aircraft during field testing

Mark GARLASCO

The last development of TLD Saint Lin

The Saint Lin factory had a record year in 2011 in term of business. Today, they are several important on going developments and products improvements on the site.

Elevating transporter TFE-3.5T:

This new equipment is mainly dedicated to load/unload containerised Airbus A320 family aircraft. This vehicle can drive like a transporter and lift like a single platform loader loads of 3.5 tons to doorsills up to 3,5 m (11,48 foot).

The selected motorization is a 3 cylinders Deutz engine (but if necessary, a 4 cylinders engine can also be used), compliant to latest emission regulations Tier 3/Tier 4i. Its hydrostatic transmission insures a very smooth approach of the aircraft. The design of this equipment aims at providing optimal maintainability, the lowest TCO as well as an easy implementation of the electric version.



TFE-3.5T

Other significant products evolutions:

✓ **WTM-900 & LTM-900-V**: these water and lavatory equipment truck-mounted on MAN chassis have been ordered for a large European airport. The interface between the operator and the unit is driven by an IFM PLC, also used on other TLD products. This design has been made in considering others chassis as well (like Mercedes Benz or Volvo) that might be requested by customers and other areas.

✓ **JET-16** : an important work has been made around all major components in order to optimize JET-16 competitiveness and to continue to gain market shares.

✓ **NBL**: to make sure that our belt loader, which has been leading the market for a long time, remains at the cutting edge of competitiveness, is a must. Therefore, the R&D investment on this product aims at optimizing the costs, while continuously improving quality and reliability.

Sébastien FABRE

The last development of TLD in China

Development of TLD in China

M. Vallon was a French aviator who had the honour of being the first person to fly an aircraft to and in China. His planes, Sommer mono- and bi-planes, caused a sensation in Shanghai when he conducted exhibition flights there in 1909.



On his last flight, he took off from a site in Jiang Wan, but the wings collapsed while he was demonstrating the plane over the central racecourse. Track management were later criticized for continuing the races after Vallon passed away.

As a minor consolation, the French gave his name to a street in their Concession after him.

Since that time, we have seen quite some progress, not only in aviation, aviators and their machines, but also in the required ground support equipment.

The first milestone of TLD presence was set in 1978, when our colleague Allen Fu and his wife Mei Hou opened the DEVTEC office in Hong-Kong. At that time, the sales to support China's ground handling went via the central CAAC (China Civil Aviation Central Agency) purchase office in Beijing. China's opening policy, promoted by Deng Xiaoping in 1979, led to a remarkable increase in air connections and resulting growth of the aviation industry; and so, only 17 years later in 1996, TLD Asia Shanghai was founded – the first foreign owned company to manufacture GSE in China.

The last development of TLD in China

TLD Asia Shanghai

Today TLD Asia Shanghai has 130 employees and comprises 3 Business Units; the factory, the Sales & Service Organisation, and the Group Sourcing Team. The Shanghai factory consists of buildings and roofed areas of 8,800 m² on a land of 14,000 m².



Assembly in the Shanghai plant covers the RANGER series of NBLs, ABS-580s as well as GPUs. Parallel to these, we also produce equipment like Catering Trucks DT 5005, Ambu-Lifts DT 5003, as well as Maintenance Platforms DT-6008 and MLP-100-T.

The Shanghai plant also supports the China Sales team by truck-mounting imported equipment (ACUs, ASUs and large GPUs from our Windsor factory) on commercial chassis.

Late in 2011 TLD Asia conducted a Value Stream Analysis and Value Stream Design process in order to optimize factory output. Applying the famous Toyota-approach, we were able to improve productivity from 15 NBL/month to 35 NBL/month, cutting down assembly times from 120 hours/NBL to around 40 hours/NBL. Workers understand themselves today as musicians in an orchestra, who have to align their team play for a harmonious concerted action – our shift leaders acting as conductors.

Considering the success of the NBL line, we are now rolling out the same process for the ABS and GPU lines. With the assistance of engineering students doing their 6-months internship with us, we collect the actual timings of assembly steps. We then combine various steps according to their sequence and length of time in order to reach similar task times.

Out of these we create daily and weekly working instructions for each worker, as well as the shift leader, so that we can follow up assembly progress “in real time” and support, in case we meet difficulties. Obviously this stream-lined timing will only bear fruit when the supply chain is also optimized. Having this in mind we have switched some of our suppliers to KANBAN and vendor managed inventories. Apart from reducing lead-time and the “missing item” problematic, we were also able to cut down on inventories and reduce working capital.

The last development of TLD in China

TLD Asia Wuxi

A second factory was founded in the Shuo Fang Aviation Industry Development Park in 2008. It is located in the city of Wuxi, 160 km north-west of Shanghai. The Wuxi plant has around 75 employees and produces the conventional push-pack tractors TMX-150 and TMX-450, baggage tractors JST, lower deck loaders TXL-838, and our Lavatory & Water trucks LSP/WSP-900 and LTM/WTM.



Unlike Shanghai, the present 3,000 m² facility is rented, it also reached maximum capacity at the end of 2011. In order to cope with the growth we are experiencing in Asia, we are therefore currently in the process of building a new green-field project under our ownership in Wuxi, located 500 m from the existing plant. The ground breaking ceremony for the new plant was held in July of last year.

The new plant will offer 9,000 m² of assembly area and some 1,700 m² of modern office space. The summer of 2012 has been pegged for completion, with a Grand Opening Event in the early autumn.

A new area of expansion for TLD operations in China!



Thomas DORN

TLD People

TLD MOSCOW OFFICE

TLD reinforces its position in Russia and the CIS (Commonwealth of Independent States)

TLD's activity in Russia and the CIS has been in constant growth over the past few years due to the strong requirements of the numerous airports of the region that entered a phase of modernization of their equipment.

TLD has been steadily conquering a growing market, from the coldest land of Yakutsk in Siberia to Moscow, the Russian capital, going also through Uzbekistan, Kazakhstan, Azerbaijan, Ukraine, Belarus, and many other countries.

Starting on the European side of the map in Moscow, TLD developed throughout the years strong business relations with the 3 airports in Moscow (Sheremetyevo, Vnukovo, Domodedovo) which are all cooperating with TLD. On the opposite side of the map, TLD has also confirmed its presence and success up to Vladivostok by winning in March 2012 an important tender.

In addition, TLD also registered great successes in CIS countries such as Ukraine. TLD recently won a contract to supply a large quantity of GSE in Kiev, Donetsk & Lvov airports for the European soccer cup in June 2012.

Finally, TLD is also currently working on a project that would reinforce its position in Russia: Sukhoi, a large civil and military Russian aircraft producer, asked TLD to work on the compatibility of our towbarless tractors for their latest commercial jet, the Superjet 100.



TLD exhibiting at the 2011 Airport Association of CIS Civil Aviation Conference



Great success for TLD: TPX-200-MT in Yakutsk, Siberia, working in extreme conditions: -60°C !

To support this growing activity, TLD has an office in Moscow. The work of this dynamic office has contributed to the success of TLD in the region for the last 15 years.



TLD People



Sergey Gershanov, the General Manager, deals with our commercial relations throughout the area. He works in close cooperation with his son, Kirill Gershanov, our Sales Manager, and together they have contributed to the considerable increase of TLD's activity in Russia and the CIS countries. As a connoisseur of the French culture

(he is famous for cooking French food sometimes better than some of us!), Sergey likes to accompany our customers to France. They are all always very satisfied with the factory visits or the technical inspections that we propose, and they always keep great memories of their visit to Paris.

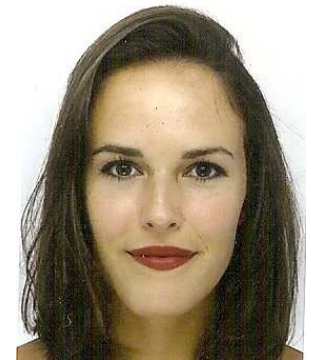
Alla Gevorkova, our trilingual – Russian, French and English - Administration Manager, considerably helps to facilitate the communication between Moscow and the other TLD offices. She is also of great help for managing the spare parts and all deliveries from Europe to Russia. Alexey Samsonian, our Sales Officer, helps with the forecast and order records and many other administrative issues. Both of them, originally from Georgia, contribute to the very international nature of TLD's teams throughout the world.

Our Service Field Technician, Dmitry Povarov, supports our customers whenever a need arises, allowing TLD to be very active and efficient on the area. He also performs all the commissioning on the region, which also contributes to the quality of TLD's customer service.



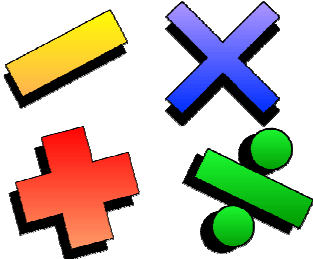
Ludovic Gaignerot, our Area Sales Manager for Central and Eastern Europe & CIS countries, works from the TLD office in Budapest and is in daily contact with the TLD team in Moscow. He makes sure, in coordination with Sergey and Kirill, that our customers are satisfied and that all proper actions are taken to expand TLD activities & customer base in the region.

Finally, Manuela Spagnol is our new recruit that has joined the TLD team on 1 January 2012. She is currently under intensive training and will be based within the TLD Moscow office reporting to Ludovic Gaignerot. She will reinforce our team and will help coordinate the work between the TLD Moscow office and all other TLD offices & factories. Manuela speaks fluent French (native), English & Russian.



Jacques ROUX

The Latest Dashboard

T L Dash board (in M€uros)			
(Year to date-end of MARCH figures)			
TLD Sales	2012-03		2012/2011 % Evolution
	in M€uros	in % of total	
GSE - AREA AMERICA (Americas, Australia, Japan, N. Zealand)	17,0	32,7%	81,9%
GSE - AREA EUROPE (Europe, Africa, middle east, India, Pakistan)	24,3	46,7%	33,5%
GSE - AREA ASIA	5,9	11,3%	<17,2>%
Other activities (Adhésives solutions)	4,8	9,3%	25,7%
TOTAL TLD SALES	52,0	100%	35,1%
			



TLD Taxibot team

